

Family 1.gf: the global fine-granularity observation family
a dataset design note — what is measured at ten kilometres and
five days or finer, and how it is stored
raw observations first; derived products only where they carry raw observations nothing
else does

blauewelt/earth — design note

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Abstract

Family 1.gf (“gf” for *global*, *fine*) is a proposed *family* — a numbered set of datasets built to one storage design and read by the programme’s forecasting model — of observation *stores* (a store is one dataset, published as a small set of arrays) covering the whole globe at a granularity of **ten kilometres or finer in space and five days or finer in time**, built on the storage contract of family 10, the programme’s granularity-aware dataset set (one registry — the JSON file that lists every store; three tiers — gridded fields (G), point measurements (P) and image tiles (T); one token — the unit the model reads, carrying a value, its offset in space and time from the place being predicted, and its footprint, the area and time span the value averages over). It extends family 10’s five point stores — 10.1’s (family 10’s first build) drifters (free-floating surface buoys), tropical moorings, ship CO₂ and 2.03 billion along-track sea-level samples (satellite altimeter heights along the ground track), and 10.2’s (its second build) fishing-effort store (E-081, the experiment that built it: the fishing fleet as a distributed sensor) — with the fine-scale measurements the programme’s four prediction goals (El Niño, the ocean carbon sink, the AMOC — the Atlantic Meridional Overturning Circulation, the Atlantic’s northward heat-carrying flow — and sea-surface temperature, SST) can each be argued to need: daily 4 km ocean colour — merged across the older sensors, and hyperspectral from PACE (NASA’s 2024 ocean-colour satellite, whose instrument resolves the whole visible spectrum and with it which kinds of phytoplankton are there) — daily 5 km and 2 km sea-surface temperature from the instruments rather than the analyses fitted to them, 6 km sea-ice concentration, 2 km swath sea level from SWOT (the Surface Water and Ocean Topography radar satellite, which maps sea level across two 50 km-wide strips), 4 km cloud-top temperature every 30 minutes, ship reports back to 1662, every non-Argo ocean profile since 1772, interior carbon bottles, biogeochemical floats, open-ocean moorings, and satellite column CO₂ (the CO₂ averaged through the whole depth of the atmosphere). Every candidate was checked against its producer on 2026-09-16: resolution, cadence, record, access, volume and licence. Three things the research settled change the design: no ocean-wind, sea-surface salinity or microwave-SST (sea-surface temperature from microwave radiometers, which see through cloud) product exists at ten kilometres, so wind is an *exception* to decide and the other two are out; the daily 4 km and 5 km fields are 10–30 % valid and 0.2–1.5 TB each even compressed, so the gridded tier gains a sharded, tiled, compressed layout for grids finer than 0.25°; and several sources have ended or changed recently (Pathfinder SST, NOAA’s 4 km infrared record, stopped at 2023, IMERG — NASA’s merged satellite precipitation — in its V07 Final run at 2025-09, and the sea-ice line of NSIDC (the US National Snow and Ice Data Center) from AMSR2 (a Japanese microwave radiometer) at 2025-08), so the family is built from the successors. The note gives the ledger, the storage design, the per-store row and byte arithmetic, a three-phase build ladder (phase A $\approx$ 0.5 TB, eight stores, keyless or hosted-runner credentials only — hosted runners are GitHub’s own build machines), the falsifier (the test that can show a store is not worth keeping) that decides whether each store earned its place, and the decisions left to Chris.

1 Scope: which measurements qualify (finer than 10 km and 5 days), and what that rule leaves out

The rule. A source qualifies for family 1.gf if it is a *global* (or global-ocean) **observation** whose spatial support is ≤ 10 km — a point measurement counts — and whose temporal support is ≤ 5 days (a pentad or

finer). A derived product is admitted only where it carries raw observations that are otherwise inaccessible (a merge of sensors whose individual level-2 files (per-orbit retrievals on the instrument’s own swath) are impractical, a bias-corrected collation whose inputs are the observations). Monthly or coarser products enter only where something the goals need would otherwise be missing, and each such exception is named as one in §6.

Why this rule, in the programme’s own terms. Family 7.2’s (the published global input tensor: every 0.25° cell, pole to pole, 1982–2024) grids are 0.25° and 1° , five-daily or monthly; family 10.1 added the point observations but no fine *fields*. E-078 (the design study that gave family 10 one schema for every granularity) measured why the grid is the wrong contract for fine data — a coarse value copied into fine cells is one measurement presented as many, a fine value averaged into a coarse cell throws away what the finer source was bought for — and E-077’s (the experiment that added ocean colour to family 7.1) ocean colour is the deliberate exception, box-averaged to 0.25° to get a channel onto the grid quickly. Family 1.gf is the fine half of that ledger, stored at its own resolution so the token’s footprint fields can say what each value averaged.

What is already held and therefore not repeated. OISST (NOAA’s daily Optimum Interpolation SST analysis) and its sea-ice fraction at 0.25° (family 7.2, g025, its 0.25° ocean-surface group); NCEP (the US National Centers for Environmental Prediction reanalysis) at 1° (g100, family 7.2’s 1° atmosphere-and-land group); GLORYS-derived (Copernicus’ ocean reanalysis) interior fields and the monthly climatology of Argo, the global array of profiling floats (rg100, the 1° monthly ocean-interior group); OC-CCI (ESA’s merged ocean-colour record) box-averaged to 0.25° (oc025, the 0.25° ocean-colour group); Argo core profiles (family 8, the Argo temperature and salinity store); drifters, tropical moorings, SOCAT (the Surface Ocean CO₂ Atlas of ship measurements) and all-mission along-track sea level (family 10.1); and the Global Fishing Watch daily 0.1° per-vessel fishing effort (family 10.2, E-081 — 11 km, so it sits just outside this family’s rule and stays where it is). Two of those are the coarse form of a store below — oc025 of oc4k (this family’s daily 4 km ocean-colour store), g025’s sea ice of `seaice_asi` (this family’s daily 6 km sea-ice store) — and both stay: the fine store is a new group, not a replacement, so a consumer that reads only tier G is unchanged.

Excluded by the rule, after checking the finest product that exists.

- **Sea-surface salinity.** Every product is ≥ 25 km on the grid and ≥ 40 km in effective resolution (RSS SMAP v6 — Remote Sensing Systems’ salinity from NASA’s Soil Moisture Active Passive satellite — “40 km”, ESA CCI SSS v5.5 — the salinity record of ESA’s Climate Change Initiative — at 25 km grid / 50 km stated, SMOS — ESA’s Soil Moisture and Ocean Salinity satellite — 25 km grid / 40–50 km footprint). Out.
- **Passive-microwave SST.** AMSR2 is a 0.25° grid from a ≈ 50 km footprint. Out; the infrared L3 products (level 3: measurements binned onto a fixed grid, not gap-filled) below are the fine-scale SST.
- **Ocean wind from scatterometers** (radars that infer wind from the roughness of the sea surface). The finest gridded product is 0.125° (≈ 13.9 km, Copernicus — the EU’s Earth-observation programme — WIND_GLO_PHY_L3_MY_012_005); the finest swath product is 12.5 km (EUMETSAT OSI SAF ASCAT L2: the Ocean and Sea Ice facility of Europe’s weather-satellite agency, processing ASCAT, the scatterometer on the MetOp satellites; CC BY 4.0, free with attribution). Neither passes. Wind is the one forcing the ocean stores cannot do without, so it is carried as **exception E1** in §6 rather than dropped silently.
- **Gap-filled analyses:** OSTIA (Met Office) and MUR (NASA JPL) SST, DUACS (the gridded altimetry of CNES and Copernicus) sea level, GLORYS, SMAP L4 (level 4: model-assimilated) soil moisture, the IMERG family’s model-like blending. They are derived; their inputs are what the stores below hold. IMERG is the one carried as an exception (E2), because the raw swaths of GPM (the Global Precipitation Measurement mission) are 40 TB and cover the globe only every few days.

2 What fine granularity buys each goal — why El Niño, the carbon sink, the Atlantic overturning and SST need detail finer than 10 km and 5 days

The four goals judge every addition (Chris, 2026-09-13). The mechanism per goal, so that the falsifier in §7 (the keep-or-drop test per store) has something concrete to test:

goal	what happens at ≤ 10 km and ≤ 5 d that a 0.25° pentad cannot see	the stores that see it
El Niño	westerly wind bursts (days, hundreds of km), the convection that drives them (cloud tops cooling by tens of kelvin in hours), oceanic Kelvin waves (equatorial waves that carry a bump of sea level eastward) as a sea-level ridge crossing the Pacific in two months, tropical instability waves at 1,000 km wavelength with 10 km fronts	irtb (infrared cloud-top temperature from weather satellites every 30 min), swot (SWOT’s 2 km sea-level swaths) and the held slatrack (family 10.1’s satellite sea level along the ground track), sst_cci05 (daily 5 km satellite SST), icoads (ship and buoy weather reports: the pre-1979 record of Pacific SST and pressure), oc4k (equatorial upwelling shows in chlorophyll first)
carbon sink	blooms at 4 km over days; the biological pump’s timing; surface $f\text{CO}_2$ (the fugacity, or effective partial pressure, of CO_2 in seawater) varies at 10 km scales along a ship track (held in socat , family 10.1’s ship CO_2 store); the interior’s stored carbon and the atmospheric gradient the flux works against	oc4k , bgcargo (biogeochemical floats: oxygen, nitrate, pH at depth), glodap (ship bottle samples: interior DIC — dissolved inorganic carbon —, alkalinity, and ages from CFCs, chlorofluorocarbon tracers), xco2 (satellite column CO_2 : the atmosphere’s side of the flux)
AMOC	deep convection in the Labrador and Irminger seas is set by the ice edge and by heat loss over days; boundary currents are 50–100 km wide with 10 km fronts; the density field that geostrophy reads comes from profiles, most of them not Argo before 2005	seaice_asi (6 km ice edge), swot (2 km sea level across the Gulf Stream and the boundary currents), wod (the World Ocean Database of ship and instrument profiles: the pre-Argo hydrography, XBT — expendable bathythermograph — lines), oceansites (open-ocean moorings) and deep_arrays (the instruments of the four AMOC monitoring arrays) — moored velocity and density time series — and sst_cci05
SST	fronts are 2–10 km wide; diurnal warming is an hourly signal that the foundation-temperature analyses (analyses of the water just below the daily warm layer) remove by design; coastal upwelling is a 10 km strip; the analyses the model has seen so far were <i>fitted</i> to these measurements, and a model that has only seen the fit cannot learn where the fit fails	sst_cci05 and sst_acspo02 (daily 2 km satellite SST; both are level-3 measurements), icoads (ship and buoy SST since 1662, the record OISST and ERSST — NOAA’s Extended Reconstructed SST — are fitted to), swh (satellite significant wave height: wave-driven mixing)

Table 1: The mechanism each goal has at fine scale, and the stores that measure it.

3 The ledger — every candidate source, checked against its producer

Every row was checked against the producer on 2026-09-16 (product pages, the NASA Common Metadata Repository, live directory and S3 (Amazon’s cloud object storage) listings, the Copernicus Marine STAC — SpatioTemporal Asset Catalog, the standard in which producers list their files). “Raw” means level-2 or level-3 measurements binned or collated without gap-filling; “derived” means an analysis. Volumes marked *native* are the producer’s files; *stored* is this note’s estimate in the layout of §4, and §5 gives the arithmetic. A dagger (†) marks a number that could not be verified against the source and is an estimate.

store	source (producer)	native res.	cadence	record	raw?	access	native vol.	licence
oc4k	<i>daily 4 km ocean colour: chlorophyll and water clarity.</i> ESA OC-CCI v6.0 merged L3S (level 3, super-collated across sensors), 4 km geographic 4320 × 8640: chlor.a (chlorophyll), kd.490 (light attenuation), Rrs (remote-sensing reflectance, the water-leaving colour; 6 bands), per-sensor nobs (observation count) (PML, Plymouth Marine Laboratory)	1/24° (4.6 km)	daily; also 5-day, 8-day	1997-09-04 → 2026-06-30, quarterly updates	raw L3S	PML THREDDS / FTP, CEDA (the UK's environmental data archive); no account	daily chlor.a 84–99 MB → ≈ 1 TB; all products 1.7 GB/day	ESA CCI free and open, cite
sst_cci05	<i>daily 5 km sea-surface temperature, per satellite sensor.</i> ESA SST CCI v3.0 L3C (level 3, collated per sensor), day and night, AVHRR (the infrared radiometer on NOAA's and Europe's MetOp weather satellites) NOAA-06.19 + MetOp-A/B and SLSTR-A/B (Sentinel-3's radiometer); 0.05° 3600 × 7200 (Reading / Met Office)	0.05° (5.6 km)	daily, 2 passes	AVHRR 1979-07 → 2024; SLSTR 2016-06 →	raw L3C	CEDA HTTPS / OPeNDAP; no account (the mirror on CDS, the Copernicus Climate Data Store, needs one)	AVHRR L3C 4 TB in 72.4 k files; SLSTR 272 GB	CC BY 4.0
sst_acspo02	<i>daily 2 km sea-surface temperature merged across sensors.</i> NOAA ACSPO (its SST processing system) L3S-LEO (level 3 super-collated, low-Earth-orbit) daily v2.81, super-collated VIIRS + AVHRR + MODIS (US imaging radiometers), 0.02° 9000 × 18000 (NOAA STAR, its satellite research centre)	0.02° (2.2 km)	daily; AM and PM variants	2000-02-24 → present –2 months	raw L3S	PO.DAAC (NASA's ocean data centre; Earthdata login, NASA's free account); NOAA CoastWatch's ERDDAP data server without	408 MB/day → 3.9 TB	NOAA open
seaice_asi	<i>daily 6 km sea-ice cover from microwave radiometers.</i> U. Bremen ASI (ARTIST sea-ice algorithm) v5.4 sea-ice concentration from AMSR-E / AMSR2 (Japanese microwave radiometers) 89 GHz (AMSR3 preliminary from 2026-06); polar-stereographic 6.25 km (1792 × 1216 N, 1328 × 1264 S) and 3.125 km	6.25 or 3.125 km	daily	2002-06-01 → 2011-10-04; 2012-07 → 2026-09-15	raw L3	HTTPS; no account	6.25 km ≈ 1.5 MB/day; AMSR2 record 8 GB (29 GB at 3.125)	“free for scientific use”, cite; redistribution terms to confirm
swot	<i>2 km sea-surface height across two wide radar swaths.</i> SWOT L2 LR SSH (level 2, low-rate sea-surface height) Expert (version D), KaRIn (its Ka-band radar interferometer) 2 km grid on two 50 km swaths; 584 passes (292 orbits) per 20.86-day cycle (JPL / CNES)	2 km	per pass	2023-07-26 → (cycle 056 on 2026-09-13)	raw L2	PO.DAAC, Earthdata login, S3 us-west-2	32 MB/pass, NASA open 18.8 GB/cycle → ≈ 1 TB	NOAA open

store	source (producer)	native res.	cadence	record	raw?	access	native vol.	licence
swh	<i>wave height along satellite altimeter tracks.</i> ESA CCI Sea State v4 L3 along-track significant wave height, all missions Topex → Sentinel-6A (the radar altimeter satellites, 1992 to today), 1 Hz (one value per second), as Copernicus WAVE_GLO_PHY_SWH_L3_MY_014_005 (one daily multi-mission file)	≈ 7 km along track	1 Hz	1991-08-03 → 2023-12-31	raw L3	Copernicus Marine (account, as slatrack)	12–16 MB/day → 150–190 GB	Copernicus Marine licence
irtb	<i>4 km cloud-top temperature every half hour from weather satellites.</i> NCEP/CPC (the US Climate Prediction Center) globally merged geostationary infrared brightness temperature, GES DISC (NASA's Earth-science data centre) GPM_MERGIR v1 : 4 km, 9896 × 3298, 60°S–60°N, int16	4 km	30 min (two per hourly file)	1998-01 → now (−24 h); 251,616 files, none missing	raw merge	GES DISC HTTPS / OPeNDAP / S3, Earthdata login	≈ 16 MB per half hour → 6.96 TB	NASA open
icoads	<i>ship and buoy weather reports since 1662.</i> ICOADS (the International Comprehensive Ocean-Atmosphere Data Set) Release 3.0 (1662–2014) + R3.0.3 near-real-time (2015 →), IMMA1 (its record format): SST, air temperature, pressure, wind, humidity, cloud, waves (NOAA NCEI, the US national environmental data centre / NCAR, the National Center for Atmospheric Research)	point	per report	1662 → 2026-08	raw	NCEI HTTPS, no account (NCAR needs one)	45 GB + 117 GB gzip; ≈ 455 M reports to 2014 [†] + ≈ 10 ⁸ /yr	CC BY 4.0 (GDEX, NCAR's data exchange) / NOAA open
wod	<i>every ocean profile except Argo floats' (ships, instruments, animals, gliders).</i> World Ocean Database 2023 + quarterly updates (Feb 2026): 18.63 M casts — OSD (bottle casts) 3.26 M, CTD (conductivity-temperature-depth) 1.13 M, MBT (mechanical bathythermograph) 2.43 M, XBT (expendable) 2.36 M, MRB (moored buoy) 1.28 M, DRB (drifting buoy) 0.27 M, PFL (profiling float: Argo, excluded here) 2.75 M, UOR (undulating recorder) 0.13 M, APB (instrumented animal) 2.06 M, GLD (glider) 2.97 M; T, S, O ₂ , nutrients, pH, alkalinity, CFCs, chlorophyll (NOAA NCEI)	point profile	per cast	1772 → 2026	raw	NCEI ragged netCDF per instrument-year, no account	167.5 GB; 121 GB without PFL	CC0 (public domain)

store	source (producer)	native res.	cadence	record	raw?	access	native vol.	licence
glodap	<i>ship bottle samples of carbon, nutrients and oxygen at depth.</i> GLODAPv3 (the Global Ocean Data Analysis Project, July 2026) bottle data: 1,498,120 samples, 1,181 cruises; DIC 541 k, alkalinity 501 k, pH 332 k, O ₂ 1.30 M, nitrate 1.16 M, CFC-12 411 k (NCEI OCADS, its ocean-carbon archive, accession 0315582)	point bottle	per cruise	1972–2023	raw, adjusted	NCEI HTTPS, no account	1.0 GB CSV	CC BY 4.0
bgcargoc	<i>profiling floats that also measure oxygen, nitrate, pH and chlorophyll.</i> BGC-Argo (biogeochemical Argo) synthetic profiles (Sprof, the merged per-float files): 2,963 floats, 407,999 profiles; DOXY (oxygen) 392 k, CHLA (chlorophyll) 195 k, BBP700 (particle backscatter at 700 nm) 192 k, NITRATE 123 k, pH 102 k (Argo GDAC, the global data assembly centre)	point profile	≈ 10 d	2002 →	raw	GDAC HTTPS/FTP, no account	≈ 15 GB	CC BY 4.0
oceansites	<i>fixed open-ocean moorings, hourly or finer.</i> OceanSITES GDAC moored time series, the 52 non-TAO sites (the 67 of TAO, the tropical Pacific mooring array, are held in <i>gtmba</i> , family 10.1’s tropical-mooring store); salinity at 103, currents at 44, O ₂ at 16 sites; hourly or better (NDBC, NOAA’s buoy centre / Ifremer, France’s ocean institute)	point mooring	≤ 1 h	1980s → (six sites reporting in 2025)	raw	NDBC / Ifremer HTTPS, no account	26 GB without the 162 GB ALOHA cabled observatory (a sea-floor station north of Hawaii)	OceanSITES open policy
xco2	<i>satellite CO₂ averaged < 3 km² through the air column.</i> OCO-2 L2 Lite XCO2 (column-averaged CO ₂) v11.2r (+ v11.3r from 2024) and OCO-3 v11r (NASA’s Orbiting Carbon Observatories): < 3 km ² soundings, 70–120 k good per day (GES DISC)		per sounding	2014-06 →; 2019-08 →	retrieval	GES DISC, Earthdata login	231 + 53 GB; 117 GB	NASA open

store	source (producer)	native res.	cadence	record	raw?	access	native vol.	licence
deep_arrays	<i>instrument records of the four Atlantic overturning arrays.</i> instrument-level data of RAPID/MOCHA 26.5°N (BODC, the British Oceanographic Data Centre), OSNAP (the subpolar North Atlantic array; Duke), SAMBA 34.5°S (the South Atlantic array; AOML, NOAA's Atlantic laboratory), MOVE 16°N (the western tropical Atlantic array; OceanSITES): T, S, velocity, bottom pressure at instrument depth	point mooring	15 min–1 h [†]	2004 / 2014 / 2009 / 2000 →	raw	public except RAPID's BODC terms, unverified	< 10 GB each [†]	open (RAPID to confirm)

Table 2: The candidates that pass the rule, as checked on 2026-09-16.

Checked and set aside. AVHRR Pathfinder v5.3 (4km, 1981-08 → 2023-12-31): NCEI marks it “no longer updated”; the SST CCI L3C record covers the same sensors from 1979 with a licence, so Pathfinder is not needed. NSIDC AU_SI6 (the AMSR-Unified 6 km product): holds only 89 GHz brightness temperatures, no concentration, and NASA stopped updating it on 2025-09-01; AU_SI12 is 12.5 km. OSI SAF OSI-408-a (its 10 km sea-ice product): a 10 km grid from an 18 km footprint, 2016 →, CC BY 4.0 — the public-track companion for sea ice; if Bremen's terms refuse redistribution, **seaice_as1** is built on the private track and OSI SAF beside it publicly. NASA OB.DAAC (the ocean-biology data centre) single-sensor 4 km chlorophyll (MODIS-Aqua, VIIRS): each is inside OC-CCI's merge. PACE OCI — NASA's hyperspectral ocean-colour instrument, 2024-03 → — is the exception OC-CCI v6 does not yet merge, and it is built as its own store **pace4k** (§5) rather than as a future **oc4k** channel, because what it adds is a spectrum and a phytoplankton community rather than one more estimate of chlorophyll; it is served by OB.DAAC under NASA's open terms and needs the Earthdata login the family already holds. GDP (Global Drifter Program) hourly drifters v2.01.1 (197 M rows, 1987 → 2022-10): four times the rows of the held 6-hourly store, ends four years earlier, adds a diurnal SST decomposition; not now. Copernicus CORA (its quality-controlled temperature and salinity profile collection; T/S profiles, 1950 → 2025-06, ≈ 90 GB): cleaner delayed-mode QC than WOD but T/S only and under the Copernicus licence; WOD is the pick for breadth and CC0, CORA the alternative if WOD's QC (quality control) costs too much in the probe (the one-month test fetch of §7). GPM DPR/GMI (its precipitation radar and microwave imager) level-2 swaths (16.5 + 22.6 TB): the raw precipitation, but the globe is covered only every few days at 245 km swath width, so IMERG is carried instead (E2). TROPOMI (the trace-gas imager on Sentinel-5P; NO₂, CO, CH₄ at 3.5 × 5.5 km, 23 + 8.5 + 2.5 TB): passes the rule and serves none of the four goals; a fire and fossil tracer for later. GOSAT (Japan's greenhouse-gas satellite) via ACOS (NASA's retrieval of it) (10.5 km footprint): just over the line, 6.9 GB; would extend **xco2** back to 2009-04 — an exception worth taking when **xco2** is built (E3).

4 Storage: what the family-10 design already gives, and the one layout it lacks — compressed tiles for grids finer than 0.25°

Family 1.gf reuses family 10.1's contract unchanged where it fits and adds one layout where it does not.

The registry. `tensors/family1.gf/family1gf.json` on `chfrank/earth-tensors` (the project's public Hugging Face dataset repository, “the Hub”), schema version 2 (family 10.1's column layout), the same top-level keys as `family10.json` (epoch 1982-01-01, `pentad_days` 5 — a pentad is the five-day bin every time is filed under —, the `bin_rule`, the `token_schema` with `log2_fp` and `log2_dt` (the base-2 logarithms of a value's spatial and temporal support, relative to 27.83 km and 5 days), per-group `tier`, `cadence`, `C`, `channels`, `footprint`, `layout`, `prefix`, files with `sha256` (a checksum), per-year counts, QC policy, `degraded` (the list of any check a build had to relax); `C` is the number of channels). The tier letters stay G, P and T so that a consumer's dispatch is unchanged; a fine grid is tier G with `layout`: “sharded”

(§4's layout) rather than "binmajor" (one array per group ordered by five-day bin), and the reader named for tier G handles both by reading that field. Six per-group fields are new and all are load-bearing: `licence` with a `redistribution` value of `yes`, `attribution` or `no` and a `derived_works` value of `free`, `non-commercial` or `restricted`, from the licence's own clause; `distribution`, `public` or `private`, the track of §4's two-track rule, `private` whenever redistribution is `no`; `frames_per_bin` and `frame_seconds` for gridded groups; `dtype` per group (float16, a two-byte number, for measured fields, uint8 for classes and for brightness temperature stored as K−160); and `verdict`, empty until §7's falsifier has run. A sharded group also names its `tile_grid.json` and `shard_index.npy`. Family 10.2's registry — the current one, which inherits 10.1's four stores by reference and adds `fishing` (E-081) — gains a one-line `siblings` list naming this file, family 1.0.tf's (land and coast at the same scale) and family 0.9.tf's (land and coast through Google's AlphaEarth embedding), so a consumer that already dispatches on `tier` discovers both families from the one entry point it knows. Nothing in family 10.1 or 10.2 is rewritten.

Tier P (points, profiles and tracks), with one widening. Every point store below ships the nine arrays of family 10.1 at schema 2 — `bin` int16, `time_s` int32 seconds, `lat`, `lon` float32, `values` float16 $[N, C]$ in raw units, `platform` int64 (the hashed instrument identity), `qc` uint8 (quality code), `fp` float16 $[N, 2]$ (the two footprint fields), `bin_offsets` int64 (where each bin's rows start) — sorted by (`bin`, `time_s`), negative bins before 1982, longitude in $[-180, 180)$ enforced in float32, one row costing $27 + 2C$ bytes. Profiles put the vertical axis into C exactly as family 8 does for Argo: the 16 Roemmich–Gilson pressures per variable (the standard levels of the Roemmich–Gilson Argo climatology, 2.5 to 2,000 dbar), NaN where the cast did not reach. The one widening: an int32 second since 1982 reaches back only to 1913-12-13, and `icoads` begins in 1662 and `wod` in 1772. A store whose record starts before 1914 declares **schema 3**, identical except that `time_s` is int64; a row then costs $31 + 2C$ bytes, `bin` (int16, ± 449 years) is unchanged, and `ml/family10_store.py` (the store reader) gains one dtype branch. Stores inside the int32 range stay at schema 2.

Tier G-fine (gridded fields finer than 0.25°), the new layout. Family 7.2's tier G is one bin-major `.npy` (NumPy array file) per group, read by one HTTP range (a request for one byte span of a file) of one bin: 14.5 MB at 0.25° for seven channels. At 0.05° one daily frame of one channel is 51.8 MB and a day of `sst_acspo02` is 324 MB per channel; a cone (the set of past measurements the model reads around the place and time it predicts, the *anchor*) anchored at one point needs a few hundred kilometres of it, and the daily fields are 10–30 % valid, the rest cloud, night or land. A flat uncompressed array would cost 5–10 times the bytes it carries and force every anchor to read a whole globe. So groups on a grid finer than 0.25° use a **sharded, tiled, compressed** layout:

- One **shard** per (group, bin): `<prefix>/<yyyy>/<bin>.shard`. Inside it, F **frames** per bin (`frames_per_bin` = 1 for a pentad field, 5 for daily, 10 for day/night, 40 for three-hourly, 240 for half-hourly), each frame cut into **tiles** of 256×256 pixels of the group's native grid (12.8° at 0.05° ; 10.7° at $1/24^\circ$; 1,600 km on the 6.25 km polar grid), each tile all C channels as float16 in raw units, NaN (not-a-number) for unobserved, zstd-compressed (Zstandard, a fast lossless compressor).
- One **index** per group: `shard_index.npy`, int64 $[n_{\text{bins}}, F, n_{\text{tiles}}, 2]$ of (byte offset, byte length) inside the bin's shard, plus `tile_grid.json` (rows, columns, the lat/lon of every tile's corner). An empty tile (all NaN — open ocean for a land product, the tropics for sea ice) has length 0 and is never stored. Tiles are float16 unless the group's `dtype` says otherwise.
- A **read** is one HTTP range per (bin, frame, tile) the cone touches: 20–130 KB compressed for a tile with 20 % valid pixels, measured by the probe (§7) before anything is built. The bin's whole frame is `n_tiles` ranges; a cone of 1,400 km reach touches four to nine tiles at 0.05° and up to about 36 at 0.02° .
- The limits of the Hub (Hugging Face's dataset hosting) shape this: under 100 k files per repository and 10 k per folder, so shards are per bin (about 3,450 for a group reaching back to 1979) in year folders, never per tile; files under 200 GB, which no shard approaches.

The frame carries its own timestamp (`frame_seconds` in the registry: 86,400 for daily, 10,800 for three-hourly; for day/night passes the local overpass time is a channel), and the token's Δt is measured from the end of the anchor's bin to the frame's time, as tier P does. A gridded value is one token at the pixel centre with $\log_2 \text{fp} = \log_2(\text{pixel km}/27.83)$ and $\log_2 \text{dt} = \log_2(\text{frame days}/5)$; nothing is copied into neighbouring pixels. On the floor: family 10.1 writes $\log_2 f_p = -4$ for a point instrument, and this note keeps that as the *label* for “no areal support”; a pixel with a real support carries its true value even below

−4 (a 1 km pixel is −4.8, a 300 m pixel −6.5), and likewise a real temporal support (an hour, −6.9) is written as it is while an instantaneous sample keeps the −4 label. E-078 §2 quoted OC-CCI’s 4 km as −2.8; the grid is $1/24^\circ = 4.64$ km, so −2.58 is the value used here.

Which tier a fine field belongs in is geometry first, bytes second. A row of tier P costs $27 + 2C$ bytes; a stored pixel of a compressed G-fine tile costs about $2C \times 1.2$ bytes and an unobserved pixel almost nothing, so on bytes alone a compressed grid beats rows at any valid fraction (an *uncompressed* grid would lose below $v = 2C/(27 + 2C)$, 7 % at $C = 1$). What decides is whether the source *has* a fixed grid: SWOT’s 2 km cells lie along two swaths whose position changes every pass, so there is no raster to tile and *swot* is a P store; a 4 km chlorophyll field at 10–15 % valid is on one grid every day and is G-fine, and a cone over the tropics then reads a few tiles instead of millions of rows. The probe measures v per source; the registry records the tier and the layout.

Tier T (image tiles) is not used by this family. Nothing global at ≤ 10 km is imagery in the tile sense — the 300 m ocean-colour and Sentinel (the EU Copernicus satellites) swaths are coastal or terrestrial and belong to family 1.0.tf (and, as a published embedding in place of the raw imagery, family 0.9.tf). This departs from E-078 §3, which placed 300 m–1 km radiometry in tier T: a *gridded level-3* product at those scales is a raster on one fixed grid and fits the sharded tier-G layout, whereas tier T is for swath imagery whose footprint moves every pass. The one candidate that approaches tier-T volume, *irtb* at 30 minutes, is handled by scope (§5) rather than by a token cache.

Two-machine builds, unchanged. Credentials — Earthdata login, Copernicus Marine (the free accounts of NASA and of the EU marine service) — live in repository secrets and reach only hosted runners; rented boxes (machines hired by the hour) are keyless and assemble from Hub parts, exactly the *slatrack* pattern, the way family 10.1 built its 67 GB sea-level store (six hosted lanes fetch and push per-year parts with *done.json*, a completion marker, written last after a restore-hash; a keyless verified box pulls, assembles by streaming, publishes with restore-verification). Every fetch lane refuses a listing that came back empty or a download that came back short and records any relaxation in *degraded*; there is no “download failed, skip” path, per Chris’s 2026-09-14 rule.

Two tracks: public and private stores. Not every source that passes the rule may be republished, and the families no longer refuse those. Chris, 2026-09-16: a dataset that forbids redistribution is used by a single party; the Hub was only ever the easiest way to import a dataset and hand it to other agents; a private dataset can be built and trained on without redistributing it. So a group’s *distribution* is **public** — the public repository *chfrank/earth-tensors*, read anonymously by anyone, the default wherever the licence allows — or **private**: the same arrays, layout and *store.json* in a private Hugging Face dataset repository (*chfrank/earth-tensors-private*, the pick, because every reader and build script is unchanged; an S3-compatible bucket with pre-signed URLs (time-limited download links) is the alternative if the private volume passes a terabyte or the Hub’s private-storage pricing bites) that only this project’s own jobs can read. The public registry lists private groups too, with their *prefix*, sizes, checks and *verdict*, so the design is public even where the bytes are not, and a consumer without a token skips them by the field rather than by a failed fetch. Four consequences. A rented box reads a private store with a fine-grained read-only token scoped to that one repository, handed to the job through a repository secret and revoked when the job ends — the one credential a box holds beyond the job’s own, admissible because it outlives nothing (*m1/CLAUDE.md* §6, the project’s security rules for rented machines; pre-signed URLs would be the keyless form). A private store is processed on a box and never kept on one: pulled for the job, deleted by the job’s hygiene, never on a stopped box’s disk, since the host has root. What may leave is the model and the numbers: a trained checkpoint, evaluation scores and aggregate statistics may be published where the licence’s derived-works clause allows (*derived_works* records it; a *restricted* source confines even the checkpoint to the private track), the data never. And a private store is a cost to reproducibility: nobody outside the project can check it, so a published result that rests on one names it, and a public source is preferred wherever one carries the same observations — which is why every CC0, CC BY (Creative Commons, attribution required) and Copernicus source in these families is on the public track.

5 The stores in numbers — rows, bytes and build phase per dataset

Row counts and stored bytes below are this note’s estimates from the verified native facts; every one is a hypothesis the one-month probe of §7 replaces with a measurement before a build is dispatched. Bytes per P row are $27 + 2C$ plus headers; G-fine bytes assume zstd at the stated valid fraction (a NaN tile compresses to nothing, a valid pixel to $\approx 2C$ bytes plus 10–20 % overhead), so **stored** $\approx$ **pixels** $\times$ **frames** $\times v \times 2C \times 1.2$.

Three arithmetic notes. **swot**’s 16 billion rows are eight times **slatrack**’s, because a swath is a grid and a nadir track is a line; at 33 bytes a row the store would be about 0.5 TB. The alternatives — keep the 250 m unsmoothed product (25 TB native) or subsample the 2 km grid — are respectively impossible and against the rule, so **swot** is built as a P store in phase B, once the Hub conversation about terabyte repositories has happened. **irtb** is the largest raw source in the ledger and its scientific value for El Niño is in the tropics, so phase B takes 30°S–30°N three-hourly (0.7 TB) and the full half-hourly 60°S–60°N record is a phase-C decision; the frame count, not the pixel count, is what the design has to hold. **sst_cci05** is built by the family’s own builder from the per-sensor L3C files, choosing the highest quality level per pixel per pass — a super-collation, which is what NOAA’s ACSPO L3S does at 0.02° from 2000; the CCI route is taken first because it reaches 1979 and is CC BY 4.0, and **sst_acspo02** follows for the 2 km detail.

Why pace4k is the 4 km daily bins and not the 1.2 km swath. PACE’s level-2 swath at 1.2 km is about a terabyte a year of cells — 2.5×10^8 ocean cells a day, a quarter of them clear — for a product whose value is the *spectrum* and the *community* rather than the last kilometre of detail, and the 4 km daily bins keep both. The swath is therefore a phase-C row, **pace1k**, to be built only if a verdict says the finer grain matters; the 4 km store is phase A. This is the one daily *binned* product admitted beside **oc4k**, and it is admitted because OC-CCI v6 does not merge PACE, so nothing else in the family carries the spectrum. Two honest notes on what it is. Its three community channels come from MOANA, which NASA labels *level 4* rather than level 3 (measured in the catalogue on 2026-09-20), so those three are a retrieval of a retrieval and are stored with the rest but flagged in **qc**; and its apparent visible wavelength comes from the optical product while chlorophyll and carbon come from the biogeochemical one, three files a day joined on one grid by the builder. Marine species records are not here: they ride **gbif**, family 1.0.tf’s store of dated, located occurrence records, which is global and holds the sea as well as the land.

store	tier	channels (C)	footprint ($\log_2 f_p, \log_2 f_t$)	rows or frames	stored	goals	phase
seaice_asi	G-fine	sea ice. 1: sic (%) on two polar grids as two sub-groups N/S	(−2.15, −2.32)	3.86 M px × 8,700 d; $v \approx 0.4$ of the polar grids	≈ 30 GB	AMOC, SST	A
oc4k	G-fine	ocean colour. 3: log_chl, kd_490, nob; +6 Rrs bands in phase B	(−2.58, −2.32)	37.3 M px × 10,500 d; $v \approx 0.10$ –0.15	0.3–0.4 TB ($C = 3$); ≈ 1.1 TB with Rrs	carbon, SST, El Niño	A
pace4k	G-fine	hyperspectral colour and the phytoplankton community. 7: chlor_a (chlorophyll), avw (apparent visible wavelength — the single colour the whole spectrum averages to), poc (particulate organic carbon), carbon_phyto (phytoplankton carbon), and MOANA’s three cell concentrations (<i>Prochlorococcus</i> , <i>Synechococcus</i> , picoeukaryotes); the daily 4 km mapped files of PACE OCI v3.2, measured in NASA’s catalogue on 2026-09-20	(−2.80, −2.32)	37.3 M px × ≈ 930 d; $v \approx 0.10$ –0.15	≈ 50 GB [†] a year, ≈ 0.13 TB [†] so far	carbon, SST, El Niño	A
icoads	P	ship reports. 8: sst, airt, slp, wind.u, wind.v, dewpt, wave_h, cloud	(−4, −4)	≈ 1.5 B [†] (455 M to 2014 + NRT, near-real-time)	≈ 75 GB	SST, El Niño, AMOC	A
wod	P	ocean profiles. 48: T, S, O ₂ at the 16 RG (Roemmich–Gilson) pressures; platform = cast id, instrument in qc bits	(−4, −4)	15.9 M casts (WOD without PFL)	2 GB	AMOC, SST	A
glodap	P	bottle chemistry. 9: depth, DIC, TA (total alkalinity), pH, O ₂ , NO ₃ , PO ₄ , SiO ₄ , CFC-12	(−4, −4)	1.50 M bottles	80 MB	carbon, AMOC	A
bgcargo	P	chemistry floats. 80: DOXY, NITRATE, PH, CHLA, BBP700 at 16 pressures	(−4, −4)	408 k profiles	80 MB	carbon	A
oceansites	P	moorings. 5: depth, T, S, u, v; one row per (site, hour, instrument)	(−4, −6.9)	$\approx 10^8$ [†]	≈ 4 GB	AMOC, El Niño, carbon	A
xco2	P	column CO ₂ . 3: xco2, xco2_unc, surface (land/water)	(−4, −4)	≈ 400 M good soundings (OCO-2) + OCO-3	≈ 18 GB	carbon	A
swot	P	swath sea level. 3: ssh_a_karin, sig0, quality	(−3.80, −4)	≈ 0.5 M px/pass × 584/cycle × 55 cycles ≈ 16 B	≈ 0.6 TB	AMOC, El Niño	B
swh	P	wave height. 3: swh, swh_denoised, swh_unc	(−2.0, −4)	≈ 2 B (the slatrack missions, 1991–2023)	≈ 70 GB	SST (mixing), read-out	B
sst_cci05	G-fine	5 km SST. 3: sst, quality, dtim; day and night as 2 frames/day, the builder super-collating sensors by best quality	(−2.32, −2.32)	25.9 M px × 2 × 16,200 d; $v \approx 0.2$	≈ 1.2 TB	SST, all	B
irtb	G-fine	cloud-top temperature. 1: tb (uint8, K–160), 30°S–30°N, 3-hourly ($F = 40$)	(−2.80, −7.9)	16.3 M px × 8/day × 10,500 d; $v = 1$, but Tb compresses $\approx 2 \times$	≈ 0.7 TB	El Niño, SST	B
precip01 (E2)	G-fine	daily satellite rain (IMERG). 2: precip, quality at 0.1°, daily	(−1.32, −2.32)	6.48 M px × 10,100 d; $v = 1$, but zeros compress	0.2–0.3 TB	El Niño, AMOC	B
deep_arrays	P	overturning arrays. 6: depth, T, S, u, v, bpr	(−4, −6.9)	10^7 – 10^8 [†]	< 5 GB	AMOC	B
sst_acspo02	G-fine	2 km SST. 2: sst, quality at 0.02°	(−3.64, −2.32)	162 M px × 9,700 d; $v \approx 0.3$	≈ 2.3 TB	SST	C
wind12 (E1)	P	scatterometer wind. 3: u10, v10, quality at 12.5 km swath cells	(−1.15, −4)	≈ 4 M cells/day/satellite × 27 yr ≈ 50 B	≈ 1.7 TB (or ≈ 50 GB as pentad 0.125° frames)	all	C
irtb full	G-fine	cloud-top temperature, whole record: 60°S–60°N, half-hourly ($F = 240$)		32.6 M px × 48 × 10,500 d	≈ 8 TB		C

Table 3: Estimated size of every store in the layout of §4, and the phase it is proposed for. Phase A totals ≈ 0.6 TB with **pace4k**; phase B ≈ 3.5 TB more; phase C is where the Hub’s “contact us for terabytes” threshold is certainly crossed.

6 The exceptions, named — three sources admitted although slightly coarser than the rule

- **E1 — scatterometer wind at 12.5 km.** Fails the rule by 25 % and is the only observation of the wind that forces every ocean store here. Proposed: admit in phase C as `wind12`, a P store of OSI SAF ASCAT L2 12.5 km swath cells (CC BY 4.0; Metop-A 2007–2014 CDR — climate data record —, Metop-B 2012 →, Metop-C 2019 →) with QuikSCAT (NASA’s earlier scatterometer, 1999–2009) from the Copernicus 0.25° line for the earlier years. The pick between the P form (1.7 TB, every cell) and pentad 0.125° frames (90 GB, a mean) is Chris’s; this note’s pick is the P form restricted to 2007 → first.
- **E2 — IMERG at 0.1° (≈ 11 km), daily.** A derived merge, gauge-calibrated in the Final run; admitted because the raw swaths cannot cover the globe daily and precipitation is the freshwater flux the AMOC goal needs. The V07 Final record ends 2025-09-30; V08 Final is expected in the second half of 2026, so the store is built when V08 lands or from V07 Final + V07 Late for the tail, with the run recorded per row in `qc`.
- **E3 — GOSAT via ACOS at 10.5 km.** Over the line by half a kilometre, 6.9 GB, and it extends the column-CO₂ record from 2014 back to 2009-04. Admitted into `xco2` with `platform` distinguishing the instrument and `fp` carrying its own footprint.
- **Monthly:** nothing. Every goal’s monthly quantities (RAPID transport, the Niño indices — SST averages over fixed Pacific boxes —, SeaFlux, a satellite estimate of the air–sea CO₂ flux) are targets or are already held coarse; no monthly source is critical to this family.

7 The build ladder, the probe, and the falsifier — the order of building, the one-month test fetch, and the test each store must pass to be kept

Phase A — nine stores, ≈ 0.6 TB, keyless or hosted-runner credentials only. `seaice.asi`, `oc4k`, `pace4k`, `icoads`, `wod`, `glodap`, `bgcargo`, `oceansites`, `xco2`, plus the registry and the `siblings` line in family 10.2’s. Seven of the nine fetch without any account; `xco2` and `pace4k` need an Earthdata login on a hosted runner (GES DISC for the first, OB.DAAC — already verified — for the second); `oc4k` is the one G-fine build large enough to need a box for assembly. Expected cost, from the 10.1 precedent (a *verified* box is a rented machine whose host the marketplace has vetted): the P stores \$0 on hosted runners; `oc4k` one verified box for a few hours, \$2–5; the whole phase under \$10 if nothing is re-fetched.

Phase B — the terabyte stores. `swot`, `sst_cci05`, `irtb` (tropics, three-hourly), `precip01`, `swh`, `deep_arrays`, and the six Rrs channels of `oc4k`. ≈ 3.5 TB; \$20–40 in box time at 10.1’s measured rates, dominated by upload; the Hub should be asked about a terabyte-scale public repository before this phase (the docs say “contact us for large repos”).

Phase C — decisions, not builds, until A and B have verdicts. `sst_acspo02`, `wind12`, the full `irtb`, and `pace1k`, PACE’s 1.2 km level-2 swath (≈ 1 TB a year) if the 4 km bins of `pace4k` turn out to be too coarse for the carbon read-out.

The probe, before any build. For each source, one month (January 2015, as E-079 — family 10’s first data pull — used) fetched on a hosted runner through the real adapter, measuring: rows per day or valid fraction per frame; compressed bytes per tile at the chosen tile size; listing completeness against the producer’s own count; fetch throughput; and for G-fine, the byte length of a cone-sized read. Family 10 learned the cost of building on the spec’s numbers — a time column 84 times coarser than the sampling, 908 rows in the wrong pentad — so **no store is dispatched on a size that has not been measured**, and the probe’s numbers replace this note’s in the registry.

The falsifier per store. A store earns its place if, at anchors where it has coverage, adding its tokens to the cone lowers the held-out masked-token loss (the model’s error at predicting tokens hidden from it) on the goal’s read-out (§2’s mapping: the Niño-3.4 — the standard El Niño index box in the central Pacific —

target for El Niño, the `socat`-scored $f\text{CO}_2$ and `oc025` colour targets for carbon, the transport measured by RAPID (the 26.5°N overturning array) and the corridor targets for AMOC, the OISST target for SST) against the same anchors, same seed, without it — three seeds (three independent training runs), because a new tier buys its own replication (`m1/CLAUDE.md` §3b, the programme’s replication rule). Parity means the coarse tensor already carried what the fine store adds at pentad cadence, and the store is then justified only as a fine *target* (E-078 §5), which is a different and cheaper claim. The verdicts are written per store into the registry’s `verdict` field so a consumer can see which fine stores the programme found to matter.

Verification, as for 10.1. Every array’s sha256 against `store.json`; N from file sizes; `bin` monotone and equal to `floor_divide(time_s, 432000)` (432,000 seconds are five days); the CSR (compressed sparse row) index reproduced; and, new for G-fine, every shard’s index entries decompressing to tiles of the declared shape with the declared NaN fraction, and per-year frame counts equal to the producer’s listing (a day absent upstream is a zero-length frame with a counter, never a silent gap).

8 Decisions for Chris — open choices, each with this note’s pick

Laid out with this note’s pick named, all reversible before phase A is dispatched.

1. **The name.** “Family 1.gf” as given; if “1” was meant as “10” (family 10.gf, the fine sibling of 10.1), the change is the prefix and the registry name, nothing else. Pick: as given.
2. **G-fine layout** — sharded tiled zstd shards with an index, versus a flat `.npz` per bin. Pick: sharded; the flat form costs 5–10× the bytes and makes every anchor read a globe.
3. **Daily frames or pentad means** for the G-fine stores. Pick: daily frames ($F = 5$); the rule asks for ≤ 5 days and the pentad mean is a derived quantity the consumer can form.
4. **Sea ice from Bremen ASI** (6.25 km, 2002 →, terms to be confirmed by email before publishing) or OSI SAF OSI-408-a (10 km grid, CC BY 4.0, 2016 →). Pick: Bremen, public if the answer is yes and private if it is no, with OSI SAF public beside it either way.
5. **SST:** CCI L3C super-collated by our builder at 0.05° first, ACSPO L3S 0.02° later. Pick: as stated.
6. **Exceptions E1–E3:** admit wind at 12.5 km (phase C), IMERG daily (phase B), GOSAT into `xc02`. Pick: admit all three; refuse SMAP-style “9 km grids from 36 km footprints” anywhere in this family.
7. **irtb scope:** tropics three-hourly first. Pick: yes.
8. **WOD (CC0) or CORA (Copernicus)** for the non-Argo profiles. Pick: WOD.
9. **The Hub:** write to Hugging Face about a terabyte-scale public dataset before phase B, and open the private repository of §4 with phase A. Pick: yes, with phase A’s measured sizes in hand.

Terms used above, in one line each

AMOC the Atlantic Meridional Overturning Circulation, the Atlantic’s northward heat-carrying flow, read out as transport at 26.5°N (RAPID). **Tier G / P / T** family 10’s three storage tiers: gridded dense fields; points, profiles and tracks; tiles of imagery. **CSR** compressed sparse row — an offsets array saying where each five-day bin’s rows start. **zstd** Zstandard, a fast lossless compressor. **STAC** SpatioTemporal Asset Catalog, the metadata standard producers publish scene listings in. **L2 / L3 / L3C / L3S / L4** processing levels: retrievals on the instrument’s swath; binned onto a grid; collated per sensor; super-collated across sensors; gap-filled analysis (derived). **Rrs** remote-sensing reflectance, the water-leaving colour per wavelength. **Kd₄₉₀** the diffuse attenuation of light at 490 nm, a water-clarity measure. **OC-CCI** ESA’s Ocean Colour Climate Change Initiative merged record (PML, Plymouth). **CEDA / THREDDS / OPeNDAP** the UK atmospheric data archive, and two protocols for reading subsets of netCDF files over HTTP. **AVHRR / SLSTR / VIIRS / MODIS** infrared radiometers on NOAA and MetOp, Sentinel-3, JPSS and Terra/Aqua satellites. **ACSPO** NOAA’s SST processing system. **ASI** the ARTIST sea-ice algorithm using 89 GHz microwave channels (AMSR-E, AMSR2, AMSR3 radiometers). **KaRIn** SWOT’s Ka-band radar interferometer that measures sea level across two 50 km swaths. **GES DISC / PO.DAAC / NCEI / AOML / NDBC / BODC** NASA’s, NOAA’s and the UK’s data centres. **ICODS / IMMA1** the ship-and-buoy marine report archive and its record format. **WOD** the World Ocean Database; its instrument codes: OSD bottle casts, CTD conductivity-temperature-depth profiles, MBT/XBT mechanical and expendable bathythermographs, MRB/DRB moored and drifting buoys, PFL profiling floats (Argo), UOR undulating recorders, APB tagged animals, GLD gliders. **GLODAP**

the Global Ocean Data Analysis Project bottle-data synthesis; **DIC** / **TA** dissolved inorganic carbon and total alkalinity; **CFC** chlorofluorocarbon tracers that date water masses. **BGC-Argo** / **Sprof** / **DOXY** / **BBP700** biogeochemical Argo floats, their merged profile files, dissolved oxygen, and particle backscatter at 700 nm. **GDAC** the Argo global data-assembly centre. **OceanSITES** / **TAO** the open-ocean moored-time-series network and the tropical Pacific mooring array (held in family 10.1's **gtmba**). **OCO-2/3** / **GOSAT** / **ACOS** NASA's and Japan's column-CO₂ satellites and NASA's retrieval of GOSAT. **TROPOMI** the trace-gas imager on Sentinel-5P. **OSI SAF** / **ASCAT** EUMETSAT's ocean-and-ice facility and the MetOp scatterometer. **OSTIA** / **MUR** / **DUACS** / **GLORYS** gap-filled SST analyses (Met Office, JPL), the gridded altimetry analysis, and the ocean reanalysis — all derived. **DPR** / **GMI** GPM's precipitation radar and microwave imager. **CORA** Copernicus' quality-controlled T/S profile collection. **PACE OCI** NASA's 2024 ocean-colour satellite and its Ocean Color Instrument, which resolves the whole visible spectrum instead of a handful of bands; **pace4k** / **pace1k** its daily 4 km bins, built in phase A, and its 1.2 km swath, a phase-C decision; **AVW** the apparent visible wavelength, the single colour a whole spectrum averages to; **MOANA** the product that turns that spectrum into concentrations of *Prochlorococcus*, *Synechococcus* and picoeukaryotes, the three groups of small phytoplankton. **GBIF** the Global Biodiversity Information Facility, whose species-occurrence records — marine ones included — are family 1.0.tf's **gbif** store. **Public** / **private track** a store anyone may read from the public repository, or one held for this project's own use because its licence forbids redistribution (§4). **E-077** / **E-079** the experiments that built ocean colour into family 7.1 and family 10's first four stores. **g025** / **g100** / **rg100** / **oc025** family 7.2's four gridded groups: 0.25° ocean surface, 1° atmosphere and land, 1° monthly interior, 0.25° ocean colour.

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